

AUTOMOTIVE DIAGNOSTIC AND ACTIVITY MONITORING

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Abstract—The Automotive Diagnostic and Activity Monitoring (ADAM) System is an integrated solution designed to provide real-time insights into the health and behavior of four-wheeled vehicles. Acting as both a diagnostic tool and a forensic black box, ADAM harmonizes multiple sensing modules—including a Global Positioning System (GPS), a gyroscope and accelerometer (MPU6050), and a vibration sensor (SW-420)—to observe vehicular dynamics. At its core, the system interfaces with the vehicle's Electronic Control Unit (ECU) via the On-Board Diagnostics (OBD) port, enabling deep access to engine parameters and fault codes. Sensor data is continuously collected, securely stored on an SD card, and synchronized with a PostgreSQL database. This rich dataset fuels a suite of machine learning models tailored for four key objectives: (1) Engine Health Monitoring, (2) Fuel Economy Analysis, (3) Driver Behavior Profiling, and (4) Predictive Maintenance. Insights derived from these models are visually rendered through an intuitive mobile application that presents the information via dynamic graphs and charts. To further enhance user engagement, the system incorporates an interactive AI-powered chatbot capable of responding to user queries and guiding necessary actions. Together, these components form a cohesive ecosystem—bridging vehicle intelligence with user accessibility, and paving the way for smarter, safer, and more informed driving experiences.

Index Terms—Raspberry pi 5, On Board Diagnostic(OBD), Global Positioning System(GPS), ECU(Electronic Control Unit)

I. INTRODUCTION

Over the last couple of years, vehicle monitoring has been a necessary measure to make vehicles and drivers safe, cut down on emissions to tackle climate change, and provide optimal performance. The significance of protecting human life and the necessity to obtain genuine information at the time of accidents for forensic analysis reinforce the importance of efficient monitoring systems. To serve this purpose, we created a vehicle monitoring logger on the basis of On-Board Diagnostics (OBD-II) port. The OBD-II port is a 16-pin standardized connector found in every contemporary vehicle, developed in the United States Of America in 1996, by the Society of Automotive Engineers (SAE). It was imposed

to enable regulation of vehicle emission so as to ensure Environmental Protection Agency(EPA) standards are met. It is a diagnostic port used to monitor real-time readings of vehicle sensors and the Diagnostic Trouble Codes (DTCs) from the Electronic Control Unit (ECU), making it a valuable resource for monitoring vehicle performance. Ecu is the main controller of the vehicle where all controls and actions are done.

OBD II port supports communication protocols such as ISO15765-4 (CAN-BUS), ISO14230-4 (KWP2000), ISO 9141-2, SAE J1850 VPW, and SAE J1850 PWM. This study is centered on the ISO 15765-4 (CAN-BUS) protocol that has been required on vehicles manufactured from 1996. The message request is sent in a hexadecimal to the protocols and it gets interpreted according to one of the five OBD II protocols and sends back a hexadecimal response. OBD II uses two types of codes to request ECU data that are Diagnostic Trouble Code (DTCs) and Parameter Identifiers (PIDs). DTCs are used to diagnose malfunctions in various subsystems of the vehicle and PIDs are used to measure real-time parameters.

II. LITERATURE REVIEW:

In recent years, there have been advancements in design and development of On-board Diagnostic and also there are machine learning algorithms for predicting engine performance, fuel consumption and anomaly detection. In this section, we review the literature on design and development of OBD boards and developed machine learning algorithms. Additionally, we identify gaps in recent literature.

A. Design and Development of On-Boards Diagnostic circuits

The study [1] contributes to the design of On-Board Diagnostic systems that provide real-time information about a vehicle, such as engine speed, coolant temperature, pressure, and Diagnostic Trouble Codes (DTCs). Data Collection of OBD Using a Keaz128 microcontroller. This paper [2] designs

an Diagnostic system that has various data collection sources like gps,imu,camera and OBD port and stores locally for every 100 ms. Using BeagleBoard-XM with TI OMAP3530 processor logs the data. This paper [3] designs of Diagnostic system that collect obd data using ELM327 microcontroller along with gps .The data is transmitted to remote server and stored in sql database and transmit data using Carambola2 WiFi module.

B. Review of Machine Learning Algorithms

This study [4] is a detailed survey of machine learning methods used in engine performance, control, and fault diagnosis. It emphasizes the employment of Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forests (RF), and neuro-fuzzy models such as ANFIS for improving fuel economy and emission control. The review also discusses the use of AI in engine control with a special focus on reinforcement learning (RL) and deep reinforcement learning (DRL) for enhanced adaptability, efficiency, and real-time decision-making in sophisticated engine operations. This paper [5] proposes an ML-based algorithm for automatic fault diagnosis in mobile networks using Self-Organizing Networks (SONs). It combines Softmax NN and SVM Models to carry out multiclass classification while diagnosing network faults. Along with KPIs and performance management counters (PMCs) for feature extraction. This unsupervised approach handles both single and multiple faults, offering a robust and adaptable solution for modern mobile networks. This paper [6] examines the use of artificial neural networks for the detection and classification of faults in internal combustion engines. It uses feedforward multilayer perceptron ANNs and training algorithms like backpropagation, Levenberg-Marquardt, quasi-Newton, extended Kalman filter, and a new Smooth Variable Structure Filter (SVSF). The study tested a method to detect engine faults without needing to know their exact location. Using vibration data from a V8 engine, converted into the crank angle domain, the system identified issues like missing bearings, chain tensioners and piston noise. Out of all the methods, SVSF, achieved 97% accuracy and avoided common training errors. It can detect and classify faults in under 30 minutes, offering a fast, low-cost solution for engine diagnostics. This paper [7] presents OBD-SecureAlert, a data-driven system that detects anomalies in vehicle behavior using CAN bus data. Since CAN buses lack built-in security measures, it is open to cyberattacks that will compromise the safety of automobiles. To overcome this situation the system uses Hidden Markov Model (HMM) to monitor and detect deviations of normal vehicle behaviour, emphasizing on detection rather than prevention. It collects data such as Speed and RPM, trains it using a sliding window and flags the data sets based on predefined threshold limits as anomalies or safety/cybersecurity threats. Attack scenarios are simulated, like injecting anomalous data into CAN bus in order to detect anomalies.

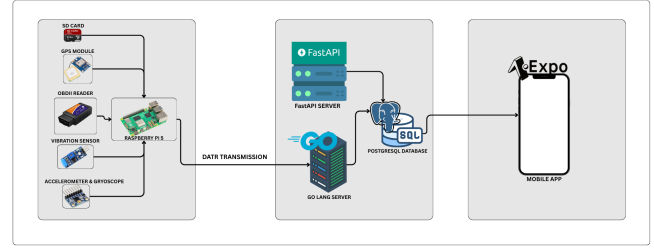


Fig. 1. System architecture of the proposed Automotive Diagnostic and Activity Monitoring (ADAM) system.

III. METHODOLOGY

A. System Architecture overview The proposed vehicle diagnostic and driver behaviour analysis system integrates various components of hardware and software to collect real time data collection for analysis and user analysis. The architecture is designed to collect vehicle information via obd II port, transmit it wirelessly to cloud server for processing and provides a visual representation of processed data through a mobile application.

The system consists of 4 primary components: Hardware Implementation Data processing using machine learning models cross platform mobile application A block diagram representing the overall system architecture and data flow from vehicle sensors to end user presentation is shown in figure 1.1

B. Hardware Implementation We have utilized an obd-II interface device compatible with standard protocols such as CAN(ISO 15765-4) and K-Line (ISO 9141-2). The device communicates with a raspberry pi 5 (a single-board computer), which is used for data collection along with other sensors like neo 6m gps module, sw 420 vibration sensor, mpu 6050 gyroscope/acceleration module. The data which is collected is stored in micro SD which can be used for forensic analysis.

C. Data Processing using machine learning model we have implemented four machine learning models that predict based on data are: driver behaviour analysis, fuel consumption, predictive maintenance, engine health performance

D. Cross platform mobile application The mobile application was developed using Expo (a cross platform framework), provides a dashboard for real time status, historical trends on fuel efficiency, driver behaviour and maintenance logs.

IV. DISCUSSION

The Automotive Diagnostic and Activity Monitoring (ADAM) System is an integrated solution designed to provide real-time insights into the health and behavior of four-wheeled vehicles. Acting as both a diagnostic tool and a forensic black box, ADAM harmonizes multiple sensing modules—including a Global Positioning System (GPS), a gyroscope and accelerometer (MPU6050), and a vibration sensor (SW-420)—to

observe vehicular dynamics. At its core, the system interfaces with the vehicle's Electronic Control Unit (ECU) via the On-Board Diagnostics (OBD) port, enabling deep access to engine parameters and fault codes. Sensor data is continuously collected, securely stored on an SD card, and synchronized with a PostgreSQL database. This rich dataset fuels a suite of machine learning models tailored for four key objectives: (1) Engine Health Monitoring, (2) Fuel Economy Analysis, (3) Driver Behavior Profiling, and (4) Predictive Maintenance. Insights derived from these models are visually rendered through an intuitive mobile application that presents the information via dynamic graphs and charts. To further enhance user engagement, the system incorporates an interactive AI-powered chatbot capable of responding to user queries and guiding necessary actions. Together, these components form a cohesive ecosystem—bridging vehicle intelligence with user accessibility, and paving the way for smarter, safer, and more informed driving experiences.

V. RESULTS

The Automotive Diagnostic and Activity Monitoring (ADAM) System is an integrated solution designed to provide real-time insights into the health and behavior of four-wheeled vehicles. Acting as both a diagnostic tool and a forensic black box, ADAM harmonizes multiple sensing modules—including a Global Positioning System (GPS), a gyroscope and accelerometer (MPU6050), and a vibration sensor (SW-420)—to observe vehicular dynamics. At its core, the system interfaces with the vehicle's Electronic Control Unit (ECU) via the On-Board Diagnostics (OBD) port, enabling deep access to engine parameters and fault codes. Sensor data is continuously collected, securely stored on an SD card, and synchronized with a PostgreSQL database. This rich dataset fuels a suite of machine learning models tailored for four key objectives: (1) Engine Health Monitoring, (2) Fuel Economy Analysis, (3) Driver Behavior Profiling, and (4) Predictive Maintenance. Insights derived from these models are visually rendered through an intuitive mobile application that presents the information via dynamic graphs and charts. To further enhance user engagement, the system incorporates an interactive AI-powered chatbot capable of responding to user queries and guiding necessary actions. Together, these components form a cohesive ecosystem—bridging vehicle intelligence with user accessibility, and paving the way for smarter, safer, and more informed driving experiences.

VI. CONCLUSION

The Automotive Diagnostic and Activity Monitoring (ADAM) System presents comprehensive, scalable and intelligent diagnostics solutions. It Integrates multiple hardware modules, machine learning models and a user-friendly mobile interface to offer real-time vehicle diagnostics, driver behavior analysis, and predictive maintenance. By combining OBD-II with additional sensor inputs and processing them using the Raspberry pi 5 it collects and stores data on the cloud and

provides insights through a mobile app with multiple functionalities including an AI chatbot. It not only aids in proactive maintenance, improved fuel efficiency and personalized user experience but also contributes to safer and greener driving.

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